Section 2: Mathematical Foundations

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1. Modular Arithmetic: Find x for the following:

* $x = 457 \pmod{13}$

$$\begin{array}{r|l} 457 & 13 \\ 67 & 35 \\ 2 & \end{array}$$

$x = 2 \pmod{13}$

* $x = -25 \pmod 7$

$$\begin{array}{r|l} 25 & 7 \\ 4 & 3 \end{array}$$

$-25 \equiv -4 \pmod 7$

$-4 + 7 = 3$

$x = 3 \pmod 7$

2. The Euclidean Algorithm:

- Calculate $\gcd(1160, 480)$ using the Euclidean Algorithm

$1160 = 480 \cdot 2 + 200$

$480 = 200 \cdot 2 + 80$

$200 = 80 \cdot 2 + 40$

$80 = 40 \cdot 2 + 0$

$\gcd(1160, 480) = 40$

- Use the Extended Euclidean Algorithm to find x, y such that $1160x + 480y = \gcd(1160, 480)$

$40 = 200 - 80 \cdot 2$ ①

$80 = 480 - 200 \cdot 2$ ②

$40 = 200 - 2(480 - 200 \cdot 2)$

$40 = 200 - 2 \cdot 480 + 4 \cdot 200$

$40 = 5 \cdot 200 - 2 \cdot 480$

$200 = 1160 - 480 \cdot 2$

$40 = 5(1160 - 2 \cdot 480) - 2 \cdot 480$

$40 = 5 \cdot 1160 - 10 \cdot 480 - 2 \cdot 480$

$40 = 5 \cdot 1160 - 12 \cdot 480$

$x = 5 \quad y = -12$

$1160(5) + 480(-12) = 40$

3. Euler's Totient Function:

- Calculate $\phi(21)$ and $\phi(17)$

$21 = 7 \times 3$

$\phi(n) = n\left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right)$

$\phi(21) = 21\left(1 - \frac{1}{7}\right)\left(1 - \frac{1}{3}\right)$

$\phi(21) = \overset{3}{\cancel{21}}\left(\frac{\overset{2}{\cancel{6}}}{\underset{1}{\cancel{7}}}\right)\left(\frac{2}{\underset{1}{\cancel{3}}}\right)$

$\phi(21) = 12$

17 es primo

$\phi(17) = 17 - 1$

$\phi(17) = 16$

- Calculate $\phi(360)$ using prime factorization

$360 = 36 \cdot 10 = (2^2 \cdot 3^2)(2 \cdot 5) = 2^3 \cdot 3^2 \cdot 5$

$\phi(360) = 360\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right)$

$\phi(360) = \overset{\cancel{120}\,24}{\cancel{360}}\left(\frac{1}{\cancel{2}}\right)\left(\frac{\cancel{2}}{\cancel{3}}\right)\left(\frac{4}{\cancel{5}}\right)$

$\phi(360) = 96$

4. Information Theory: A source produces A, B, C, D with probabilities $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{4}$, $P(C) = \frac{1}{8}$, and $P(D) = \frac{1}{8}$.

- Calculate the Shannon Entropy $H(X)$.

$H(X) = -\sum_i P(x_i) \log_2 P(x_i)$

Para 1/2: $\log_2\left(\frac{1}{2}\right) = -1$

$-\frac{1}{2}(-1) = \frac{1}{2}$

Para 1/4: $\log_2\left(\frac{1}{4}\right) = -2$

$-\frac{1}{4}(-2) = \frac{1}{2}$

Para 1/8: $\log_2\left(\frac{1}{8}\right) = -3$

$$-\frac{1}{8}(-3) = \frac{3}{8}$$

$$H(x) = \frac{1}{2} + \frac{1}{2} + \frac{3}{8} + \frac{3}{8}$$

$$H(x) = 1 + \frac{3}{4}$$

$$H(x) = 7/4 = 1.75$$

5. Bayes Theorem: An instrusion Detection System (IDS) is designed to flag a "Brute Force" attack.

- The Probability of an actual attack is $P(A) = 0.001$
- If there is an attack, the IDS correctly flags it with a probability of 0.99 (True Positive)
- If there is no attack, the IDS incorrectly flags it with a probability of 0.01 (False Positive)
- Question: If the IDS flags an event, what is the probability that an actual attack is occurring?

A = Ataque real

$P(A) = 0.001$

$P(\neg A) = 0.999$

$P(F|A) = 0.99$

$P(F|\neg A) = 0.01$

$$P(A|F) = \frac{P(F|A) \cdot P(A)}{P(F|A) \cdot P(A) + P(F|\neg A) \cdot P(\neg A)}$$

$$P(A|F) = \frac{0.99 \cdot 0.001}{0.99 \cdot 0.001 + 0.01 \cdot 0.999}$$

$$P(A|F) = \frac{0.00099}{0.01098}$$

$$P(A|F) = 0.09016 = 9.016\%$$